



## WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**

WPS No. **RNC-WS-002**

Date **23-Feb-13**

Revision No. **1**

Supporting PQR No.(s) **RNC-PQR-002**

Date **23-Feb-13**

Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding (GTAW)**

(Automatic, **Manual**, Machine, or Semi-Auto.)

### JOINT (QW-402)

Joint Design **Single V**

Backing (Yes) ☐ (No) ☒ X

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal

☐ Nonmetallic ☐ Other

Notes

**See Details Attachment sheet Joint Sketch**

### BASE METALS (QW-403)

P-No. **1** Group No. **1 or 2** to P-No. **1**

Group No. **1 or 2**

Material Specification Type and Grade **A106 Gr.B, A234 WPB Equivalents**

TO **A106 Gr.B, A234 WPB Equivalents**

Chem. Analysis and Mech. Prop. **-**

Thickness Range :

Base Metal : Groove **1.6 mm - 17.12 mm.** Fillet **All**

Pipe Dia. Range : Groove **All** Fillet **All**

Other

### FILLER METALS (QW-404)

#### GTAW

Spec. No. (SFA)

**5.18**

AWS No. (Class)

**ER 70S-G**

F-No.

**6**

A-No.

**1**

Size of Filler Metals

**Ø 2.0 & 2.4 mm**

Deposited Weld Metals

**8.56 mm.**

**Thickness Range :**

Groove **17.12 mm Max.**

Fillet **All**

Electrode-Flux (Class)

**-**

Flux Trade Name

**-**

Consumable Insert

**-**

Other

**Kobelco TGS-50 or Equivalents**

### POSITION (QW-405)

Position(s) of Groove **ALL**

Welding Progression Up ☒ X Down ☐

Position(s) of Fillet **ALL**

### POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **N/A**

Time Range **N/A**

Notes **T>20 mm. heat treatment 593-649 Holding 1 hr**

### PREHEAT (QW-406)

Preheat Temp. Min. **T < 25 mm. = 10°C**

Interpass Temp. Max. **T ≥ 25 mm. = 79°C**

Preheat Maintenance **N/A**

Notes

### GAS (QW-408)

#### Percent Composition

|           | Gas(es) | (Mixture) | Flow Rate     |
|-----------|---------|-----------|---------------|
| Shielding | Argon   | 99.99%    | 10-20 Ltr/Min |
| Trailing  | -       | -         | -             |
| Backing   | -       | -         | -             |

WPS No.

RNC-WS-002

Rev.

1

### ELECTRICAL CHARACTERISTICS (QW-409)

Current ☐ AC ☒ DC Polarity EN  
Amps (Range) 105-119 Volts (Range) 9-12  
Tungsten Electrode Size and Type Ø 2.4 mm , Thoriated ( EWTh-2 )  
(Pure Tungsten, 2% Thoriated, etc.)  
Mode of Metal Transfer for GMAW N/A  
(Spray arc, short circuiting arc, etc.)  
Electrode Wire feed speed range N/A

### TECHNIQUE (QW-410)

String or Weave Bead STRING OR WEAWE  
Orifice or Gas Cup Size NO. 4-9  
Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING  
Method of Back Gouging N/A  
Oscillation N/A  
Contact Tube to Work Distance N/A  
Multiple or Single Pass (per side) MULTIPASS  
Multiple or Single Electrodes SINGLE ELECTRODE  
Travel Speed (Range) 4.97-9.28 Cm/Minute  
Peening N/A  
Other N/A

| Weld Layer(s) | Process | Filler Metal |           | Current     |            | Volt Range | Travel Speed Range (Cm/min) | Other |
|---------------|---------|--------------|-----------|-------------|------------|------------|-----------------------------|-------|
|               |         | Class        | Dia. (mm) | Type Polar. | Amp. Range |            |                             |       |
| 1st           | GTAW    | ER 70S-G     | 2.4       | DC EN       | 100-105    | 9-10       | 4-11                        |       |
| 2nd           | GTAW    | ER 70S-G     | 2.4       | DC EN       | 105-108    | 9-10       | 4-11                        |       |
| Other(s)      | GTAW    | ER 70S-G     | 2.4       | DC EN       | 105-119    | 10-12      | 4-11                        |       |
|               |         |              |           |             |            |            |                             |       |
|               |         |              |           |             |            |            |                             |       |
|               |         |              |           |             |            |            |                             |       |
|               |         |              |           |             |            |            |                             |       |

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of ASME-IX, 2010 latest Revision.

Prepared by:

Mr. Chatchawan S.

Name

Signed

23/02/13

Date

Reviewed by:

ภูชิน สีหอมไชย
Phuchin Seehomchai

Name

Signed

11/03/13

Date

Approved by:

ภูชิน สีหอมไชย
Phuchin Seehomchai

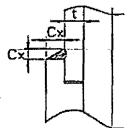
Name

Signed

11/03/13

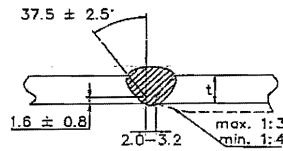
Date

# ATTACHED FIGURE JOINT SKETCH



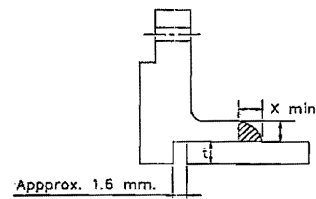
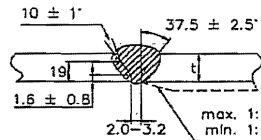
Cx (min.) = 1.25t but not less than 3.2 mm.

JOINT No. A1



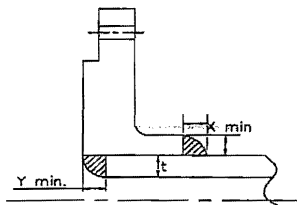
Misalignment (max.) = 1.5 mm.

JOINT No. B1 & B2



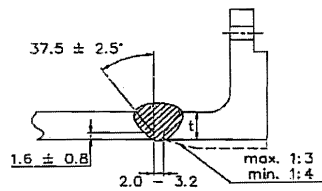
X min. = The lesser of 1.4t or thickness of the hub.

JOINT No. A2



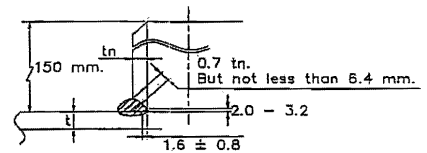
X min. = The lesser of 1.4t or thickness of the hub.  
Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



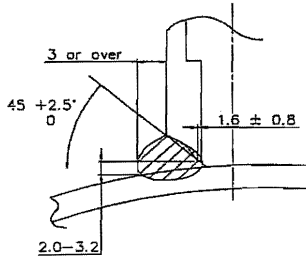
Misalignment (max.) = 1.5 mm.

JOINT No. B3

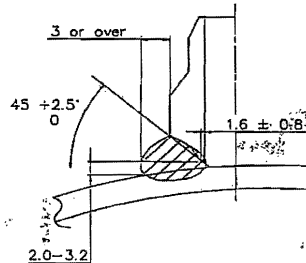


Misalignment (max.) = The lesser of 3.2 mm. or 0.5tn.

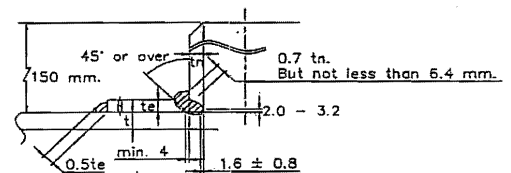
JOINT No. B4



JOINT No. C1

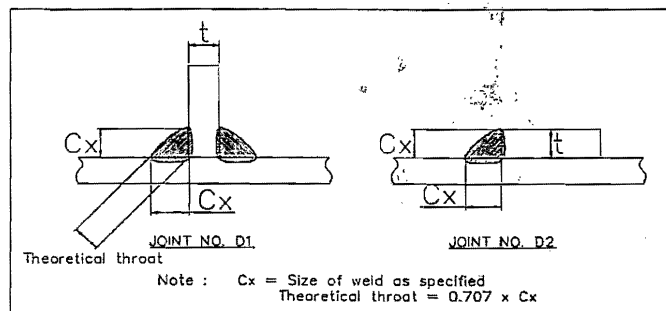


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or 0.5tn.

JOINT No. B5





# Procedure Qualification Record (PQR)

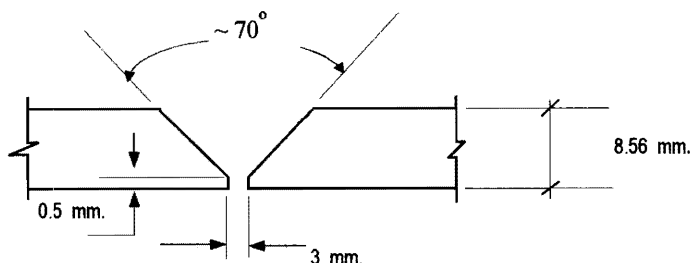
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-002

|                                          |                                          |        |                  |
|------------------------------------------|------------------------------------------|--------|------------------|
| Company Name :                           | RENEWCOILS ENGINEERING & SUPPLY CO.,LTD. |        |                  |
| PQR Record Number :                      | RNC-PQR-002                              | Date : | 23 February 2013 |
| WPS Number :                             | RNC-WS-002                               |        |                  |
| Welding Process (es) :                   | Gas Tungsten Arc Welding (GTAW)          |        |                  |
| Type (Manual, Semi-auto, or Automatic) : | Manual                                   |        |                  |

## JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

## WELDING VARIABLES

|                                         |                |         |           |                                            |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
|-----------------------------------------|----------------|---------|-----------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|--|----------|---------|-----------|-----------|-------|--------|-----------|----------|---|---|---|---------|---|---|---|
| <b>BASE METALS (QW-403)</b>             |                |         |           | <b>POSTWELD HEAT TREATMENT (QW-407)</b>    |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Material Spec.                          | A 106          | To      | A 106     | Temperature                                | N/A                                                                                                                                                                                                                                                                                                                          |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Type or Grade                           | Gr.B           | To      | Gr.B      | Hold Time                                  | N/A                                                                                                                                                                                                                                                                                                                          |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| P. No.                                  | 1              | To      | 1         | Others                                     | <b>GAS (QW-408)</b><br><br>Percent Composition <table><tr><td></td><td>Gas (es)</td><td>Mixture</td><td>Flow Rate</td></tr><tr><td>Shielding</td><td>Argon</td><td>99.99%</td><td>11 L/Min.</td></tr><tr><td>Trailing</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Backing</td><td>-</td><td>-</td><td>-</td></tr></table> |         |         |  | Gas (es) | Mixture | Flow Rate | Shielding | Argon | 99.99% | 11 L/Min. | Trailing | - | - | - | Backing | - | - | - |
|                                         | Gas (es)       | Mixture | Flow Rate |                                            |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Shielding                               | Argon          | 99.99%  | 11 L/Min. |                                            |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Trailing                                | -              | -       | -         |                                            |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Backing                                 | -              | -       | -         |                                            |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Thickness of Test Coupon                | 8.56 mm.       |         |           |                                            |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Diameter of Test Coupon                 | Ø 4"           |         |           |                                            |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Others                                  |                |         |           |                                            |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| <b>FILLER METALS (QW-404)</b>           |                |         |           | <b>ELECTRICAL CHARACTERISTICS (QW-409)</b> |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| SFA Specification                       | A 5.18         |         |           | Current                                    | DC                                                                                                                                                                                                                                                                                                                           |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| AWS Classification                      | ER70S-G        |         |           | Polarity                                   | EN                                                                                                                                                                                                                                                                                                                           |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Filler Metal F-No.                      | 6              |         |           | Amperage                                   | 105 - 119                                                                                                                                                                                                                                                                                                                    | Voltage | 10 - 10 |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Weld Metal Analysis A-No.               | 1              |         |           | Tungsten Electrode Size                    | ø 2.4 mm. 2% thoriated.EWTh-2                                                                                                                                                                                                                                                                                                |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Size of Filler Metal                    | Ø 2.4 mm.      |         |           | Others                                     | N/A                                                                                                                                                                                                                                                                                                                          |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Others                                  | Kobelco TGS-50 |         |           | <b>TECHNIQUE (QW-410)</b>                  |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Weld Metal Thickness                    | -              |         |           | Travel Speed                               | 4.97 - 9.28 cm/min.                                                                                                                                                                                                                                                                                                          |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| <b>POSITION (QW-405)</b>                |                |         |           | String or Weave Bead                       | Both                                                                                                                                                                                                                                                                                                                         |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Position of Groove                      | 6G             |         |           | Oscillation                                | N/A                                                                                                                                                                                                                                                                                                                          |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Progression of Welding (Uphill or Down) | Up Hill        |         |           | Multiple or single Pass                    | Multiple Pass                                                                                                                                                                                                                                                                                                                |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Others                                  | N/A            |         |           | Multiple or Single Electrodes              | Single Electrode                                                                                                                                                                                                                                                                                                             |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| <b>PREHEAT (QW-406)</b>                 |                |         |           | Others                                     | N/A                                                                                                                                                                                                                                                                                                                          |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Preheat Temperature                     | 33° C          |         |           |                                            |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Interpass Temperature (max.)            | 179° C         |         |           |                                            |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |
| Others                                  | N/A            |         |           |                                            |                                                                                                                                                                                                                                                                                                                              |         |         |  |          |         |           |           |       |        |           |          |   |   |   |         |   |   |   |

# Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- RNC-PQR-002

Tensile Test Report No.TS-13-02-083

| Specimen   | Width<br>(mm) | Thickness<br>(mm) | Area<br>(mm <sup>2</sup> ) | Ultimate<br>Total Load (KN) | Ultimate<br>Strength (N/mm <sup>2</sup> ) | Type of<br>Failure & Loc. |
|------------|---------------|-------------------|----------------------------|-----------------------------|-------------------------------------------|---------------------------|
| 544/13 TR1 | 19.13         | 8.14              | 155.72                     | 85.10                       | 546.51                                    | OUT OF WELD               |
| 544/13 TR2 | 19.11         | 8.28              | 158.230                    | 85.80                       | 542.22                                    | OUT OF WELD               |

Guided Bend Tests (QW-160) Report No.BD-13-02-032

| Type and Figure Number | Type of Bend         | Indication            | Results  |
|------------------------|----------------------|-----------------------|----------|
| 544/13 BF1             | Transverse face bend | No Open Discontinuity | Accepted |
| 544/13 BF2             | Transverse face bend | No Open Discontinuity | Accepted |
| 544/13 BR1             | Transverse root bend | No Open Discontinuity | Accepted |
| 544/13 BR2             | Transverse root bend | No Open Discontinuity | Accepted |

Toughness Tests (QW-170)

| Specimen<br>Number | Notch<br>Location | Notch<br>Type | Test<br>Temp. | Impact<br>Values | Lateral Expansion |      | Drop Weight |          |
|--------------------|-------------------|---------------|---------------|------------------|-------------------|------|-------------|----------|
|                    |                   |               |               |                  | % Shear           | Mils | Break       | No Break |
| N/A                | N/A               | N/A           | N/A           | N/A              | N/A               | N/A  | N/A         | N/A      |

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A

Macroetch Test Results N/A

## Other Tests

Type of Test Radiographic examination results Accepted Report No.WQT-006

Hardness Testing Report No. HV-13-02-014

Macrostructure Report No. MA-13-02-048

Impact Testing Report No.IM-13-02-091

Welders Name Mr.Prathom M.

Welder Number W-009

Tests Conducted By Mr.Keerati Sa-ngoen

Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :



Inspected by :

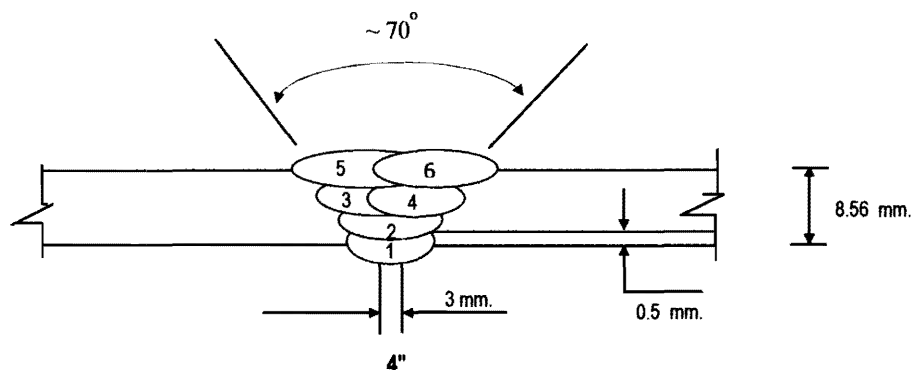
Certified by :

Date :



: 23 February 2013

**Welding Sequences for PQR-RNC-002 (Butt Joint)**



| Weld Bead No. | Process | Class Filter Metal | Wire Size (mm) | Arc Current (Amps) | Arc Voltage (Volts) | Pot. AC/DC+ | String or  | Wire Feed | Rate of Travel (cm/min) | Heat Input (KJ/cm) | Gas Shielding Flow Rate (l/min) | Gas Backing Flow Rate (l/min) |
|---------------|---------|--------------------|----------------|--------------------|---------------------|-------------|------------|-----------|-------------------------|--------------------|---------------------------------|-------------------------------|
|               |         |                    |                |                    |                     |             | Weave Bead | Speed     |                         |                    |                                 |                               |
| 1             | GTAW    | ER70S-G            | 2.4            | 105                | 10                  | DCEN        | -          | -         | 4.97                    | 12.68              | 11                              | -                             |
| 2             | GTAW    | ER70S-G            | 2.4            | 108                | 10                  | DCEN        | -          | -         | 9.28                    | 6.99               | 11                              | -                             |
| 3             | GTAW    | ER70S-G            | 2.4            | 106                | 10                  | DCEN        | -          | -         | 7.62                    | 8.35               | 11                              | -                             |
| 4             | GTAW    | ER70S-G            | 2.4            | 119                | 10                  | DCEN        | -          | -         | 7.36                    | 9.71               | 11                              | -                             |
| 5             | GTAW    | ER70S-G            | 2.4            | 118                | 10                  | DCEN        | -          | -         | 6.24                    | 11.35              | 11                              | -                             |
| 6             | GTAW    | ER70S-G            | 2.4            | 116                | 10                  | DCEN        | -          | -         | 6.10                    | 11.41              | 11                              | -                             |
|               |         |                    |                |                    |                     |             |            |           |                         |                    |                                 |                               |



# PAE TECHNICAL SERVICE CO.,LTD

## NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

### WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

|                                                       |  |                          |  |                          |  |                    |  |
|-------------------------------------------------------|--|--------------------------|--|--------------------------|--|--------------------|--|
| Client :<br>Renew Coils Engineering & Supply Co.,Ltd. |  | Project Name : PQR       |  | Location : RNC Work Shop |  | Report No. PQR-018 |  |
|                                                       |  | Customer Order No. : N/A |  | WPS No. : RNC-WPS-002    |  | Date : 13-Feb-13   |  |

| Welder No. | Welder Name   | Welding Process | Test Position | Applicable Standard | Material  |                    | Electrode Specification | Results  |          |
|------------|---------------|-----------------|---------------|---------------------|-----------|--------------------|-------------------------|----------|----------|
|            |               |                 |               |                     | Spec      | Size               |                         | Visual   | RT       |
| W-009      | MR.PRATHOM M. | GTAW            | 6G            | ASME IX             | A106 Gr.B | ø 4" Thk. 8.56 mm. | A5.18                   | ACCEPTED | ACCEPTED |
|            |               |                 |               |                     |           |                    |                         |          |          |

**Joint Details**

| Layer (S) | Process | Filler Metal |            | Current       |     | Volt | Travel Speed cm/min | Interpas Temp °C | Heat Input (KJ / cm) | Shielding |                   | Backing |                   |
|-----------|---------|--------------|------------|---------------|-----|------|---------------------|------------------|----------------------|-----------|-------------------|---------|-------------------|
|           |         | Class        | Dia. (mm.) | Type Polarity | Amp |      |                     |                  |                      | GAS       | Flow Rate (l/min) | GAS     | Flow Rate (l/min) |
| 1         | GTAW    | ER70S-G      | 2.4        | DCEN          | 105 | 10   | 4.97                | 33               | 12.68                | Argon     | 11                | -       | -                 |
| 2         | GTAW    | ER70S-G      | 2.4        | DCEN          | 108 | 10   | 9.28                | 91               | 6.99                 | Argon     | 11                | -       | -                 |
| 3         | GTAW    | ER70S-G      | 2.4        | DCEN          | 106 | 10   | 7.62                | 103              | 8.35                 | Argon     | 11                | -       | -                 |
| 4         | GTAW    | ER70S-G      | 2.4        | DCEN          | 119 | 10   | 7.36                | 179              | 9.71                 | Argon     | 11                | -       | -                 |
| 5         | GTAW    | ER70S-G      | 2.4        | DCEN          | 118 | 10   | 6.24                | 150              | 11.35                | Argon     | 11                | -       | -                 |
| 6         | GTAW    | ER70S-G      | 2.4        | DCEN          | 116 | 10   | 6.10                | 153              | 11.41                | Argon     | 11                | -       | -                 |
|           |         |              |            |               |     |      |                     |                  |                      |           |                   |         |                   |
|           |         |              |            |               |     |      |                     |                  |                      |           |                   |         |                   |
|           |         |              |            |               |     |      |                     |                  |                      |           |                   |         |                   |
|           |         |              |            |               |     |      |                     |                  |                      |           |                   |         |                   |

PAE Inspection By

SIGNATURE :

NAME :

DATE :

Mr.Wichit K.

14-Feb-2013

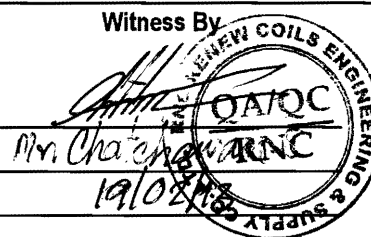


Witness By

SIGNATURE :

NAME :

DATE :



Approved By

SIGNATURE :

NAME :

DATE :

Phuchin Seehomchai

Phuchin Seehomchai



LPN PLATE MILL

## Testing Laboratory LPN Plate Mill Public Company Limited

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29779

## Tension Testing Report

Customer : RENEWCOILS ENGINEERING &amp; SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-02-083

Revision No. 0

|                        |                       |                           |                                           |
|------------------------|-----------------------|---------------------------|-------------------------------------------|
| Test Product           | WPS No. RNC-WPS-002   | Material Characterization | Pipe                                      |
| Project Name           | WELDER SUPPLY AND PQR | Test Method               | ASTM A370-10 (Prepared by ASME IX : 2010) |
| Project No.            | N/A                   | Standard of Test          | ASME Sec. IX : 2010 edition               |
| Material Specification | A106 Gr.B             | Type of Test              | Reduced Section Tension                   |
| Dimension              | Ø 4" (8.56 mm Thk.)   | Equipment's Capacity      | 100 TONS                                  |
| Date Received          | 19 February 2013      | Equipment / Serial No.    | INSTRON / 1000DXR1414                     |
| Date Tested            | 20 February 2013      | Ambient Temperature       | 23 ± 5 °C                                 |

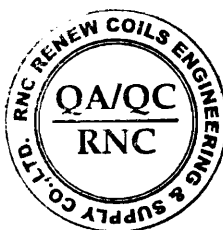
| Item No. | Test No.   | Specimen Dimension |                   |               |                            |                      | Yield        |                                  | Ultimate     |                                  | Elongation (%) | Remark      |
|----------|------------|--------------------|-------------------|---------------|----------------------------|----------------------|--------------|----------------------------------|--------------|----------------------------------|----------------|-------------|
|          |            | Diameter<br>(mm)   | Thickness<br>(mm) | Width<br>(mm) | Area<br>(mm <sup>2</sup> ) | Gauge Length<br>(mm) | Load<br>(kN) | Strength<br>(N/mm <sup>2</sup> ) | Load<br>(kN) | Strength<br>(N/mm <sup>2</sup> ) |                |             |
| 1        | 544/13 TR1 | N/A                | 8.14              | 19.13         | 155.72                     | N/A                  | N/A          | N/A                              | 85.10        | 546.51                           | N/A            | OUT OF WELD |
| 2        | 544/13 TR2 | N/A                | 8.28              | 19.11         | 158.23                     | N/A                  | N/A          | N/A                              | 85.80        | 542.22                           | N/A            | OUT OF WELD |
|          |            |                    |                   |               |                            |                      |              |                                  |              |                                  |                |             |
|          |            |                    |                   |               |                            |                      |              |                                  |              |                                  |                |             |
|          |            |                    |                   |               |                            |                      |              |                                  |              |                                  |                |             |
|          |            |                    |                   |               |                            |                      |              |                                  |              |                                  |                |             |
|          |            |                    |                   |               |                            |                      |              |                                  |              |                                  |                |             |
|          |            |                    |                   |               |                            |                      |              |                                  |              |                                  |                |             |
|          |            |                    |                   |               |                            |                      |              |                                  |              |                                  |                |             |
|          |            |                    |                   |               |                            |                      |              |                                  |              |                                  |                |             |

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009



Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date ...22... / ...02... / ...13...

Approved by :

( Mr. Payang Pienwonghat )

Deputy Managing Director

Date ...22... / ...02... / ...13...

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29785

## Bend Testing Report

Customer : RENEWCOILS ENGINEERING &amp; SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-02-032

Revision No. 0

|                        |                       |                           |                             |
|------------------------|-----------------------|---------------------------|-----------------------------|
| Test Product           | WPS No. RNC-WPS-002   | Material Characterization | Pipe                        |
| Project Name           | WELDER SUPPLY AND PQR | Test Method               | ASME Sec. IX : 2010 edition |
| Project No.            | N/A                   | Standart Of Test          | ASME Sec. IX : 2010 edition |
| Material Specification | A106 Gr.B             | Type of Jig               | Guided-Bend Roller Jig      |
| Dimension              | Ø 4" (8.56 mm Thk.)   | Diameter of Meudrel       | 32 mm                       |
| Date Received          | 19 February 2013      | Bend Angle                | 180 Degree.                 |
| Date Tested            | 20 February 2013      | Equipment / Serial No.    | SHIMADZU UH200A/32098926    |

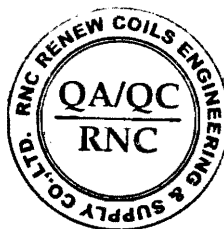
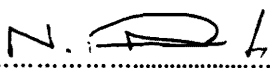
| Item No. | Test No.   | Type of Bend         | Size of Specimen<br>(ThickxWidthxLength) | Indication            | Location of Indication | Remark |
|----------|------------|----------------------|------------------------------------------|-----------------------|------------------------|--------|
| 1        | 544/13 BF1 | Transverse face bend | 8.14x19.14x160.73 mm                     | No Open Discontinuity | N/A                    | -      |
| 2        | 544/13 BF2 | Transverse face bend | 8.11x19.15x160.77 mm                     | No Open Discontinuity | N/A                    | -      |
| 3        | 544/13 BR1 | Transverse root bend | 8.14x19.17x160.46 mm                     | No Open Discontinuity | N/A                    | -      |
| 4        | 544/13 BR2 | Transverse root bend | 8.68x19.14x160.55 mm                     | No Open Discontinuity | N/A                    | -      |
|          |            |                      |                                          |                       |                        |        |
|          |            |                      |                                          |                       |                        |        |
|          |            |                      |                                          |                       |                        |        |
|          |            |                      |                                          |                       |                        |        |
|          |            |                      |                                          |                       |                        |        |

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009

Reported by : 

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : 

(Mr.Pavaret Freedawinhat)

Deputy Managing Director

Date 22 / 02 / 13

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## Impact Testing Report

Customer : RENEWCOILS ENGINEERING &amp; SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-02-091

Revision No. 0

|                        |                       |                           |                                              |
|------------------------|-----------------------|---------------------------|----------------------------------------------|
| Test Product           | WPS No. RNC-WPS-002   | Material Characterization | Pipe                                         |
| Project Name           | WELDER SUPPLY AND PQR | Test Method               | ASTM A370-10 and ASTM E23M-07a <sup>e1</sup> |
| Project No.            | N/A                   | Standard of Test          | ASME Sec. IX : 2010 edition                  |
| Material Specification | A106 Gr.B             | Type of Test              | Charpy V-Notch Impact                        |
| Dimension              | Ø 4" (8.56 mm Thk.)   | Equipment / Serial No.    | Tinius Olsen / 221811                        |
| Received Date          | 19 February 2013      | Equipment 's Capacity     | 542 Joules                                   |
| Tested Date            | 20 February 2013      | Ambient Temperature       | 23 ± 5 °C                                    |

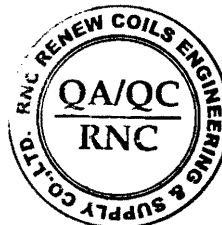
| Item No. | Test No. | Test Temperature (°C) | Specimen Dimension |            |             | Results                  |                  | Location of Test | Remark |
|----------|----------|-----------------------|--------------------|------------|-------------|--------------------------|------------------|------------------|--------|
|          |          |                       | Thickness (mm)     | Width (mm) | Length (mm) | Energy Absorbed (Joules) | Average (Joules) |                  |        |
| 1        | 544/13-1 | -46°C                 | 6.713              | 10.006     | 54.88       | 14.68                    | 26.33            | Weld Metal       | -      |
| 2        | 544/13-2 | -46°C                 | 6.713              | 10.004     | 54.87       | 51.75                    |                  |                  |        |
| 3        | 544/13-3 | -46°C                 | 6.713              | 10.008     | 54.84       | 12.57                    |                  |                  |        |
| 4        | 544/13-1 | -46°C                 | 6.713              | 10.007     | 54.91       | 80.46                    | 92.08            | Base Metal       | -      |
| 5        | 544/13-2 | -46°C                 | 6.714              | 10.008     | 54.86       | 100.51                   |                  |                  |        |
| 6        | 544/13-3 | -46°C                 | 6.713              | 10.010     | 54.90       | 95.27                    |                  |                  |        |
| 7        | 544/13-1 | -46°C                 | 6.713              | 10.008     | 54.95       | 175.12                   | 167.41           | HAZ              | -      |
| 8        | 544/13-2 | -46°C                 | 6.714              | 10.009     | 54.98       | 179.75                   |                  |                  |        |
| 9        | 544/13-3 | -46°C                 | 6.714              | 10.008     | 54.96       | 147.36                   |                  |                  |        |

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009



Reported by :

( Mr.Keerati Sa-ngoen )

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by :

( Mr.Pavaree Pradwiphat )

Deputy Managing Director

Date 22 / 02 / 13

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Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

13895

## Macrostructure Analysis Report

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

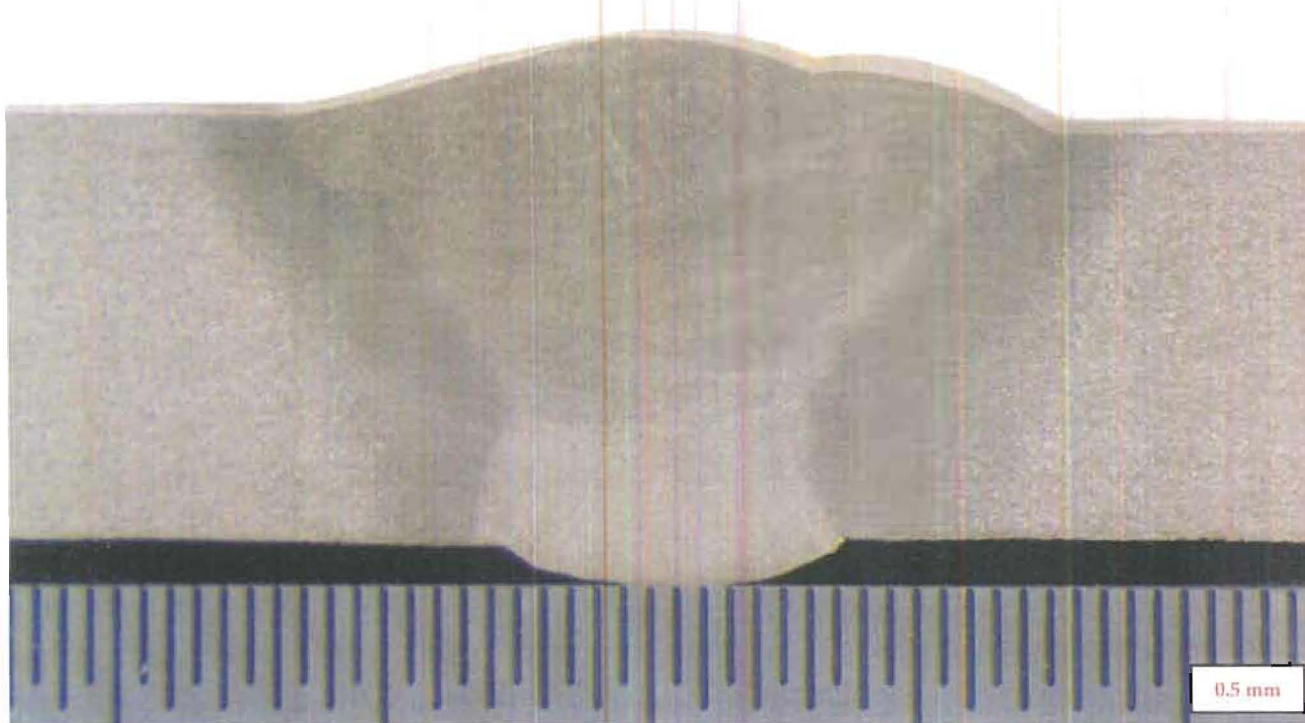
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000 Thailand.

Page 1 of 1

Report No. MA-13-02-048

Revision No.0

|                          |                        |                           |                                                 |
|--------------------------|------------------------|---------------------------|-------------------------------------------------|
| Test Product             | WPS No. RNC-WPS-002    | Material Characterization | Pipe                                            |
| Test No.                 | 544/13 MA              | Equipment used            | Microscope upto 40X (times)                     |
| Test Location            | -                      | Serial No.                | LEICA 0650854912                                |
| Material Specification   | A106 Gr.B              | Method of Preparation     | ASME Secction IX : 2010 edition                 |
| Dimension                | 4" (8.56 mm Thk.)      | Test Method               | ASME Secction IX : 2010 edition                 |
| Welding Process/Position | GTAW/6G                | Standard of Test          | ASME Secction IX : 2010 edition                 |
| Type of Joint            | Butt Jiont             | Etchant Reagent           | 25 ml HNO <sub>3</sub> + 75 ml H <sub>2</sub> O |
| Welder Name / No.        | Mr. Prathom M. / W-009 | Time of Etching           | 5 minutes                                       |
| Date Received            | 18 February 2013       | Date Tested               | 20 February 2013                                |



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

Additional details : -

Reported by :

( Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date : 22/02/13



Approved by :

( Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : 21/02/13

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## Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

13231

## Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEER &amp; SUPPLY CO.,LTD.

Report No. HV-13-02-014

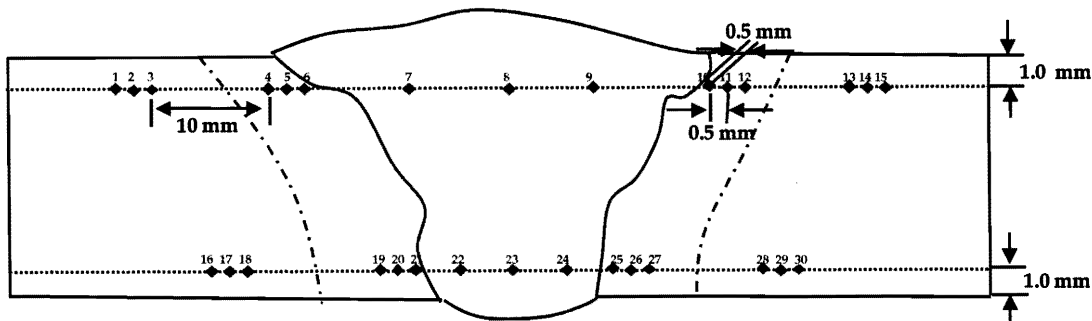
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

|                          |                       |                           |                                             |
|--------------------------|-----------------------|---------------------------|---------------------------------------------|
| Test Product             | WPS No. RNC-WPS-002   | Material Characterization | Pipe                                        |
| Test No.                 | 544/13 HN             | Test Method               | ASTM E92-82 (Reapproved 2003) <sup>E2</sup> |
| Project Name             | WELDER SUPPLY AND PQR | Standard of               | ASME Section IX : 2010 edition              |
| Project No.              | N/A                   | Type of test              | Vicker Hardness Test                        |
| Material Specification   | A106 Gr.B             | Equipment                 | Wilson Wolpert , 432 SVD (V32D219)          |
| Dimension                | 4" (8.56 mm Thk.)     | Indenter Size             | Pyramid diamond, angle of 136°              |
| Welding Process/Position | GTAW/6G               | Test Load                 | HV 10 Kgf.                                  |
| Type of Joint            | Butt Jiont            | Total Force Dwell Time    | 15 s                                        |
| Date Received            | 18 February 2013      | Cal. Block                | No. A60023 No. A62072                       |
| Date Tested              | 20 February 2013      | Ambient Temperature       | 26.0 °C                                     |

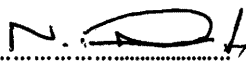
Test Location

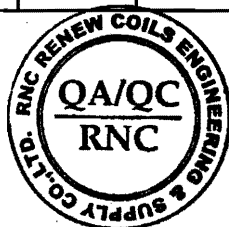
Total = 30 points



| Point No. | Location | Hardness value<br>(Scale HV ) | Point No. | Location | Hardness value<br>(Scale HV ) | Point No. | Location | Hardness value<br>(Scale ) |
|-----------|----------|-------------------------------|-----------|----------|-------------------------------|-----------|----------|----------------------------|
| 1         | BASE     | 147.5                         | 16        | BASE     | 154.8                         | *****     |          |                            |
| 2         | BASE     | 148.3                         | 17        | BASE     | 152.5                         |           |          |                            |
| 3         | BASE     | 148.5                         | 18        | BASE     | 151.8                         |           |          |                            |
| 4         | HAZ      | 166.7                         | 19        | HAZ      | 162.2                         |           |          |                            |
| 5         | HAZ      | 176.4                         | 20        | HAZ      | 165.6                         |           |          |                            |
| 6         | HAZ      | 189.2                         | 21        | HAZ      | 169.2                         |           |          |                            |
| 7         | WELD     | 196.6                         | 22        | WELD     | 155.4                         |           |          |                            |
| 8         | WELD     | 191.2                         | 23        | WELD     | 154.6                         |           |          |                            |
| 9         | WELD     | 192.2                         | 24        | WELD     | 153.9                         |           |          |                            |
| 10        | HAZ      | 188.4                         | 25        | HAZ      | 164.6                         |           |          |                            |
| 11        | HAZ      | 177.4                         | 26        | HAZ      | 166.9                         |           |          |                            |
| 12        | HAZ      | 175.0                         | 27        | HAZ      | 162.0                         |           |          |                            |
| 13        | BASE     | 151.8                         | 28        | BASE     | 152.4                         |           |          |                            |
| 14        | BASE     | 152.4                         | 29        | BASE     | 152.5                         |           |          |                            |
| 15        | BASE     | 152.6                         | 30        | BASE     | 149.8                         |           |          |                            |

Additional details : -

Reported by :   
 ( Mr.Keerati Sa-ngoen )  
 (Acting) General Manager - Quality Assurance  
 Date : 22 / 02 / 13



Approved by :   
 ( Mr.Keerati Sa-ngoen )  
 Quality Managing Director  
 Date : 22 / 02 / 13

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ArcelorMittal South Africa Limited  
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South Africa

# MATERIAL TEST CERTIFICATE SEAMLESS TUBE

Telephone +27 (0)16 450 4220  
Fax +27 (0)16 423 4906



Licensed under  
API Spec 5L and 5CT  
5L - 0319  
5CT - 0419



ISO 9001: 2008  
DE-395997 QM  
Test Certificate  
EN 10204:2004 TYPE 3.1



|                           |                                                                                                                                                                                                                              |                             |             |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------|
| Customer:                 | UR MENA LIMITED                                                                                                                                                                                                              | Customer Order/Contract No: | STH7 136520 |
| Order No:                 | 4000012782                                                                                                                                                                                                                   | Material No:                | 1000000403  |
| Certificate Reference No: | 040060162150                                                                                                                                                                                                                 | Cast/Heat No:               | 98B2332 /   |
| Product:                  | FULLY KILLED HOT FINISHED CARBON STEEL SEAMLESS TUBES                                                                                                                                                                        | Page 1 of 1                 |             |
| Specification:            | ISO3183:2007/API 5L:2007 L245/L290/B/X42 PSL1 ASTM A106.08B/A53B.07/ A530.04 ASME SA106.08B/SA53B.07/SA530.04                                                                                                                |                             |             |
| Product Marking:          | ARCELORMITTAL SA ISO3183/Spec 5L-0319 MONOGRAM 07-11 ASTM/ASME A/SA106B A/SA53B 114.300 (4.500) 8.560 ( 0.337) 6.000 (19.700) L245/B L290/X42 PSL1<br>SMLS TESTED 20.7 MPa (3000 psi) CAST NO:98B2332 PROD/O NO: T7741278210 |                             |             |

## General Information

| Quantity | Mass           | Dimensions  |           |             | Total Length  |
|----------|----------------|-------------|-----------|-------------|---------------|
|          |                | Tube OD     | Thickness | Length      |               |
| 59       | 7,901.280(kg)  | 114.300(mm) | 8.560(mm) | 6.000(m)    | 354.000(m)    |
|          | 17,419.162(lb) | 4.500(" )   | 0.337(" ) | 19.700 (ft) | 1,162.300(ft) |

| Steel making process | Final Rolling Operation                                                   |
|----------------------|---------------------------------------------------------------------------|
| Electric Arc         | Final hot rolling operation finished above 860°C and cooled in still air. |

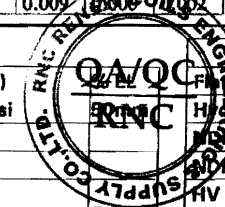
## Chemical Composition

| Element(%)    | C     | Si    | Mn    | S     | P     | Cr    | Ni    | Mo    | Cu    | V     | Al    | Ti    | Sn    | Ca     | N      | B      | Nb     | CE    | R22   | R24   |  |  |  |  |  |  |  |  |
|---------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|--------|--------|--------|-------|-------|-------|--|--|--|--|--|--|--|--|
| Minimum       | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -      | -      | -      | -      | -     | -     | -     |  |  |  |  |  |  |  |  |
| Maximum       | 0.200 | -     | 1.30  | 0.030 | 0.030 | 0.50  | 0.50  | 0.150 | 0.500 | -     | -     | -     | -     | -      | -      | -      | -      | 0.41  | 0.15  | 0.060 |  |  |  |  |  |  |  |  |
| Heat          | 0.171 | 0.23  | 0.86  | 0.003 | 0.014 | 0.12  | 0.09  | 0.038 | 0.110 | 0.005 | 0.043 | 0.001 | 0.006 | 0.0012 | 0.0089 | 0.0002 | 0.0020 | 0.36  | 0.01  | 0.007 |  |  |  |  |  |  |  |  |
| Product       | 0.171 | 0.230 | 0.860 | 0.003 | 0.014 | 0.120 | 0.090 | 0.038 | 0.110 | 0.005 | 0.043 | 0.001 | 0.006 | 0.001  | 0.009  | 0.000  | 0.002  | 0.360 | 0.008 | 0.007 |  |  |  |  |  |  |  |  |
| Product (ADD) | 0.171 | 0.230 | 0.860 | 0.003 | 0.014 | 0.120 | 0.090 | 0.038 | 0.110 | 0.005 | 0.043 | 0.001 | 0.006 | 0.001  | 0.009  | 0.000  | 0.002  | 0.360 | 0.008 | 0.007 |  |  |  |  |  |  |  |  |

## Mechanical Properties

| Specification | UTS (Rm) |       | Yield (0.5%) |       | % EL<br>50mm |
|---------------|----------|-------|--------------|-------|--------------|
|               | MPa      | psi   | MPa          | psi   |              |
| Limits        | 415      | 60000 | 290          | 42000 | 30.0         |
| Minimum       | -        | -     | -            | -     | -            |
| Maximum       | -        | -     | -            | -     | -            |
| (1) Actual    | 506      | 73389 | 369          | 53519 | 36.0         |
| (2) Actual    | 502      | 72808 | 365          | 52938 | 37.0         |
| (3) Actual    |          |       |              |       |              |
| (4) Actual    |          |       |              |       |              |

|                                    | UTS (Rm) |     | Yield (0.5%) |     |
|------------------------------------|----------|-----|--------------|-----|
|                                    | MPa      | psi | MPa          | psi |
| (5) Actual                         |          |     |              |     |
| (6) Actual                         |          |     |              |     |
| (7) Actual                         |          |     |              |     |
| Orientation & type of tensile test |          |     |              |     |
| Longitudinal, Strip                |          |     |              |     |
| Width of tensile piece (mm)        |          |     |              |     |
| 25.4mm                             |          |     |              |     |
| Orientation of impact test piece   |          |     |              |     |



## OTHER TESTS

| Passed                         |
|--------------------------------|
| 20700 kPa (3000 psi) - 5 sec   |
| PASS - ASTM E570 - 12.5% NOTCH |
| UT not required                |
| ISO 154 156                    |
| HY 10 Kg                       |
| COPIED                         |
| 04501909                       |

## Remarks:

Material in accordance with NACE MR0175:2003/ISO15156-2:2003, MR0103:2010. Dimensions to ANSI B36.10M-2004  
The material conform to the hot yield strength requirements as per ASME, Sect II, Pt D, Table Y-1, 2010  
All the material conform to the visual and dimensional requirements and is made to a suitable fine grain practice

Quality Assurance Manager: R. Bester

Date of Issue: 2011.07.29

Certified by: R

We hereby certify that the steel grade and quality level of all products are in conformity with the order and comply fully with specification requirements. No changes, amendments or additions may be made to this document. Any changes which are effected shall invalidate this certificate.

**KOBELCO**

# **INSPECTION CERTIFICATE**

BUYER'S NAME

Date of Issue 17/Feb/2011

Certificate No. KF11-002

Date of Inspection 13/Feb/2011

| TRADE DESIGNATION | SIZE (mm) | MFG. No. | APPLICABLE SPECIFICATION & CLASSIFICATION |
|-------------------|-----------|----------|-------------------------------------------|
| TG-S50            | 2.4       | F0B2704  | AWS A5.18 ER70S-C                         |

## Chemical Composition (%)

| Elements | C    | Si   | Mn   | P     | S     | Cu   | Ni   | Cr   | Mo     | V     |  |
|----------|------|------|------|-------|-------|------|------|------|--------|-------|--|
| WIRE     | 0.10 | 0.75 | 1.39 | 0.011 | 0.017 | 0.13 | 0.01 | 0.01 | < 0.01 | 0.002 |  |

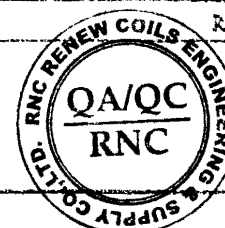
## Tensile Test of Weld Metal

## Impact Test of Weld Metal

| Yield Strength<br>at 0.2% offset | Tensile Strength | Elongation | Test Temp. | Absorbed Energy | Tested size for<br>Mechanical test: 2.4 mm<br>Tested MFG. No. of<br>Mechanical test: F9B4140 |
|----------------------------------|------------------|------------|------------|-----------------|----------------------------------------------------------------------------------------------|
| 453 MPa                          | 545 MPa          | 38 %       | -30 °C     | Avg. 257 J      |                                                                                              |

## Welding Conditions

| Welding Process | DCEN  | Shielding Gas | Ar |
|-----------------|-------|---------------|----|
| Welding Current | 135 A |               |    |
| Welding Voltage | 14 V  |               |    |



## REMARKS

We hereby certify that the test results of the above welding material are as described herein and satisfy the requirements for applicable specification.



**KOBE MIG WIRE (THAILAND) CO., LTD.**



CHIEF INSPECTOR

*[Signature]*

PROCEL